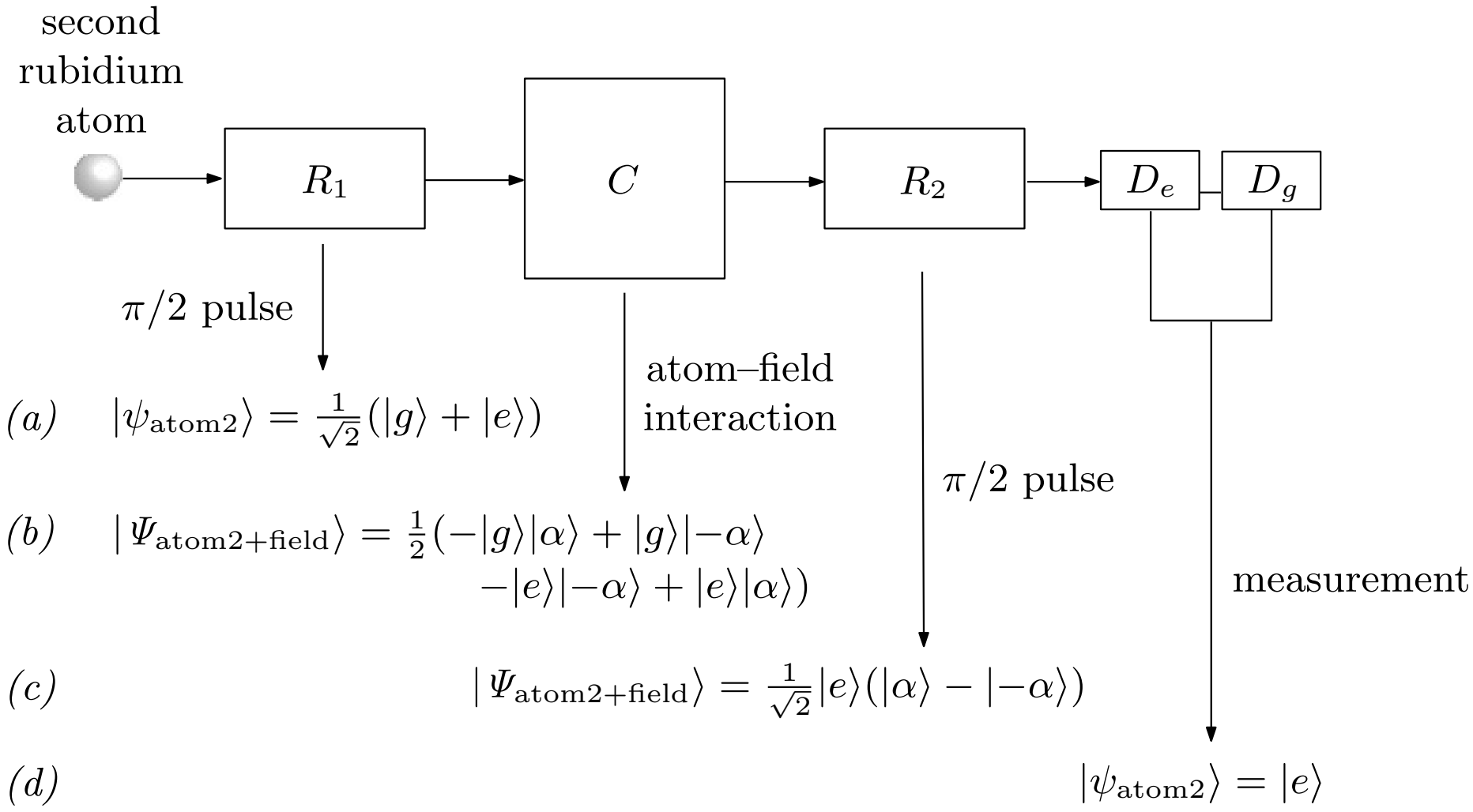

second
rubidium
atom
R_1
C
R_2
D_e
D_g
$\pi/2$ pulse
atom–field
interaction
$\pi/2$ pulse
measurement
(a) $|\psi_{\text{atom2}}\rangle = \frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)$
(b) $|\Psi_{\text{atom2+field}}\rangle = \frac{1}{2}(-|g\rangle|\alpha\rangle + |g\rangle|-\alpha\rangle$
$-|e\rangle|-\alpha\rangle + |e\rangle|\alpha\rangle)$
(c) $|\Psi_{\text{atom2+field}}\rangle = \frac{1}{\sqrt{2}}|e\rangle(|\alpha\rangle - |-\alpha\rangle)$
(d) $|\psi_{\text{atom2}}\rangle = |e\rangle$